Chapter 1 Introduction and motivation

1.1 Reasoning as a sequential allocation problem

Language models can answer mathematical and scientific questions by producing intermediate text before committing to an answer. Chain-of-thought prompting makes that intermediate work explicit [1]. The resulting sequence can contain useful calculations, checks, and repairs. It can also contain repetition, an incorrect reinterpretation of the problem, or a revision that replaces a correct candidate with an incorrect one. Additional generation therefore has an uncertain benefit and a measurable cost. A stopping policy must decide when the expected benefit of another increment justifies that cost.

The problem is especially clear when a model supplies several successive candidate answers to the same question. A trajectory may begin incorrectly, recover, and then become incorrect again. Its final answer alone conceals both the earlier opportunity to stop and the risk of stopping before a later repair. Comparing only the first and last answers similarly conceals the sequence of changes. The appropriate observational unit is a trajectory with ordered candidate answers, and the relevant decision is made using the prefix available at the moment of stopping.

This thesis studies that decision in a deliberately controlled setting. Each reasoning increment is a complete model response containing a candidate answer. The model can be asked to revise that response at the next increment. The experiments are therefore about repeated response-and-revision trajectories. They should not be interpreted as observations of every latent thought, every token of an uninterrupted internal reasoning

process, or the model's unobserved cognition. This distinction determines what a controller can interrupt and how savings must be measured.

The motivating application is economical inference: spend computation when it improves the answer enough to be worthwhile, and terminate when further computation has low expected value. Economical inference is not identical to maximizing accuracy at any cost. A policy can improve a cost-sensitive utility while reducing accuracy; conversely, a policy can increase accuracy while increasing total computation. Both outcomes must be reported. Describing either one solely as a successful stop rate would hide the actual trade-off.

1.2 Operational meaning of overthinking

Let a saved candidate be correct when it satisfies the benchmark's frozen reference grader. A repair is an incorrect-to-correct transition between successive saved candidates. A corruption is a correct-to-incorrect transition. These terms describe an observable change in the answer label. They do not assert that the model was internally uncertain or that its narrative accurately reports its reasoning.

Three phenomena must be distinguished. First, **answer corruption** means that a later candidate is wrong after an earlier candidate was correct. Second, **unproductive continuation** means that additional generation fails to produce enough improvement to offset its cost. Third, **policy regret** means that a stopping decision has lower utility than a specified comparator. Corruption can occur without large regret when it is rare or inexpensive to repair. Unproductive continuation can occur even when accuracy is constant. Regret can arise because the policy stops before a repair, not only because it continues beyond a correct answer.

Published work on overthinking motivates the possibility that longer reasoning is inefficient [8]. This thesis asks a narrower, measurable question: on the recorded model-domain panels, when does another response increment have negative net value, and how much of that information can a causal controller use? The literature supplies context; the project's measured effects come from its own frozen artifacts.

At the population level, an accuracy curve can show where the average benefit of continuation changes sign. That curve is not a per-question stopping oracle. An early difficulty assessment, answer stability, or model confidence may differentiate questions that share the same average curve. Conversely, two indistinguishable prefixes may have different unobserved continuations. A scientifically useful controller must acknowledge both possibilities rather than treating the panel's best step as a universally correct stopping time.

1.3 Research questions

This study asks how repair and corruption explain changes in graded correctness and whether the resulting gains justify computation cost. It also asks when a one-step drift criterion is an optimal stopping rule. Chapter 2 distinguishes conditional hazards from joint transition frequencies and establishes the finite-horizon solution, a sufficient persistence condition, and counterexamples.

The empirical question concerns how model family, scale, domain, temperature, token cap, and numerical precision relate to the observed trade-off. The operational question is whether a prefix-only controller makes timely decisions and whether its loop avoids further generation. Chapters 3-5 distinguish descriptive associations, controlled contrasts, controller latency, live completion tokens, and replayed counterfactual costs.

1.4 Contributions

The contribution is a protocol-specific study of accuracy-cost trade-offs in complete response-and-revision trajectories. It combines established stopping theory with controlled empirical contrasts and a causal implementation. The findings depend on the declared generation, answer-selection, grading, and cost conventions; Section 1.6 compares them with prior work.

Chapter 2 applies finite-horizon Bellman/Snell theory [11], [12] to a specified decision filtration, selected answer, and cumulative computation cost. It derives conditional repair-corruption drift and distinguishes one-step gain from optimal multi-step continuation value. It proves a sufficient persistence condition for myopic optimality and gives counterexamples without that condition. Drift-error bounds and stake comparisons have explicit assumptions. These results apply established theory; they do not prove the fitted controller optimal.

The empirical work compares matched hazard estimators and paired token-cap and precision settings, evaluates detectors with task-level grouping, and classifies policy failures. It distinguishes ranking from stopping utility and future-informed diagnostics from deployable prefix predictors. In the recorded matched contrasts, more expressive estimators and calibration can reduce utility. These negative results identify limitations of particular stopping approximations.

The analysis keeps the variable-horizon boundary and five-step detector corpora separate, attaching roster, split, sample size, and horizon to each result. A prefix-only controller with a response floor ceases future generation. Separate fitting, calibration, and evaluation tasks yield a serializable probability model without runtime reference

labels. Checks cover future-step independence, terminal behavior, token accounting, and adversarial trajectories; manifests identify executed sources and artifacts.

In both live panels, the learned rule stops every task at the minimum two-response budget. Observed savings therefore do not establish an adaptive advantage over that fixed budget. Low absolute accuracy and broad paired intervals limit accuracy-preservation claims. These negative findings accompany the working causal implementation. Comparative superiority over prior methods and a universally best reasoning horizon remain unestablished.

1.5 Evidence and reproducibility

Frozen manifests identify the selected data, label versions, and retained software evidence. They distinguish historical runtime records from the current workstation environment. Hashes establish artifact identity under the declared byte-normalization rule; they do not establish label validity, representative sampling, or bit-identical regeneration.

Historical tables retain their recorded labels. Later grader changes and reconstructed predictor targets have separate versioned results. Regression tests establish the checked behaviors, while semantic label validity requires independent adjudication. Appendix A maps the evidence and locates the electronic verification and reanalysis guide; Chapter 6 discusses these limits.

1.6 Related work and the thesis position

Chain-of-thought prompting studies how explicit intermediate reasoning can improve language-model performance [1]. Its benefit does not imply that every additional

reasoning increment is valuable. Verifier training uses completed solutions to learn candidate selection [2], while process supervision provides feedback on intermediate mathematical steps [6]. The present endpoint is the selected answer at a saved response boundary. It does not certify every preceding mathematical statement, so the project's answer-level detectors are distinct from process verifiers. These approaches establish why intermediate work and evaluation can help; the stopping question concerns the conditional value of continuing a chosen protocol after a particular prefix.

Adaptive computation predates current language-model reasoning systems. Adaptive Computation Time learns a halting unit for repeated recurrent-state updates and includes a penalty for computation in its training objective [16]. PonderNet instead models the conditional probability of halting, optimizes expected prediction loss over possible halting steps, and regularizes the halting distribution toward a prior [17]. Both learn how much internal computation to perform within a trainable architecture. Their computational increments are state updates, whereas this thesis observes complete generated responses from a fixed generator. The connection is the allocation of effort according to the evolving state; the difference is the information available to the policy and the operation whose execution it can prevent.

Confident Adaptive Language Modeling, or CALM, chooses intermediate-layer exits during autoregressive decoding and calibrates local per-token decisions against sequence-level performance constraints [18]. The present controller instead decides whether to request another complete response. Layer execution and response generation measure different resources, and a response-boundary heuristic does not inherit CALM's calibration guarantees.

Self-consistency samples several reasoning paths and aggregates their answers [7]. Adaptive-Consistency makes that sampling budget variable using agreement among the samples already generated [19]. It is a direct precedent for prefix-dependent allocation, with the prefix consisting of completed sampled solutions. The present revision trajectory instead conditions later responses on earlier generated work and uses a declared causal answer selector. Independent-path aggregation and successive revision need not have the same transition law or correctness endpoint. Agreement features can be useful in either setting, but their acquisition cost and timing must be included. In particular, thirteen completed peer trajectories cannot be treated as a free observation in a cost comparison with a single generator.

The overthinking literature motivates inefficiency from excessive test-time reasoning [8]. More recent work addresses termination within a chain of thought. Liu and Wang examine answer convergence and evaluate answer-consistency stopping, changes to end-of-reasoning signals, and a supervised stopping predictor using internal activations [20]. Their work shows that stability is a serious candidate stopping signal, rather than a new idea introduced by this thesis. Stability, however, is a statement about answer agreement; the project separately measures correctness and the possibility of a later repair. Its complete-response boundaries and observable prefix summaries also differ from intervention within a single reasoning trace or access to internal activations.

REFRAIN combines a reflective-redundancy discriminator with a sliding-window upper-confidence-bound controller that adapts stopping thresholds [21]. This provides a recent training-free alternative to a fitted probability model. A redundancy score, a stable answer, and a probability of correctness are different quantities, and each policy must specify how its score determines a continuation decision. The present confidence-and-

stability heuristic and trained drift rule are evaluated under their own fixed thresholds and response protocol; REFRAIN's reported results are contextual evidence, not estimates of this implementation's savings or accuracy.

Optimal stopping supplies the established mathematical distinction between immediate reward and conditional continuation value [11], [12]. A particularly close reasoning application is OS-Pruner, a July 2026 preprint that optimizes answer accuracy minus a token-length penalty after observing a reasoning prefix [22]. It intervenes at paragraph boundaries, elicits a final answer on termination, and trains a stopping policy using hidden-state features and precomputed rewards. It also presents a Bellman interpretation and distinguishes continuation value from a fixed correctness threshold. Thus, applying an accuracy-cost optimal-stopping objective to reasoning is already explicit in prior work. This thesis studies selected answers at complete-response boundaries and estimates one-step drift from observable summaries. Its fitted rule approximates a different target from a learned full continuation-value policy, and its finite counterexamples explain the extra conditions needed to connect myopic drift to optimal stopping.

Probability calibration is another established component. Platt's sigmoid mapping fits probabilities from classifier scores [23], and Guo et al. distinguish probability calibration from classification accuracy and evaluate post-processing methods for neural networks [24]. The project uses independently held-out task groups to fit its Platt calibrators and reports ranking and probability errors separately. A calibration fit on archived prefixes does not establish calibration under the live generator, parser, or stopping-induced state distribution. The actual live model is the instruction-tuned Qwen2.5-0.5B release [25];

identifying its model snapshot is essential because the wider archived model panels do not validate transport to every Qwen release or to larger reasoning-specialized models.

Sequential uncertainty theory addresses a further question: how can an interval or decision certificate retain its stated error control under repeated observation? Hoeffding's concentration bound concerns bounded observations under specified sampling assumptions [13]. Confidence sequences provide time-uniform coverage under the conditions of their construction [14]. Such results differ from fitting a calibrated probability score, and neither an arbitrary confidence threshold nor repeated use of an ordinary bootstrap interval inherits their guarantees. Chapter 2 states sufficient assumptions for its uncertainty results; Chapters 4 and 5 report empirical uncertainty at the declared experimental unit. This keeps a mathematical stopping certificate distinct from a descriptive interval for a measured policy contrast.

Prior work establishes learned halting, adaptive sampling, within-chain truncation, and accuracy-cost stopping. This thesis studies complete-response revision under a specified generation, grading, and cost protocol, comparing repair-corruption rules with fixed budgets. Its contribution is protocol-specific derivations and controlled findings, including failures. Prior methods were not reproduced under a shared protocol, so comparative superiority remains untested.

1.7 Thesis organization

Chapter 2 develops the stopping theory and Chapter 3 specifies the experiments. Chapters 4-5 report empirical and online results; Chapter 6 interprets them. Appendices supply the evidence map, claim scope, paired uncertainty, supporting analyses and diagrams.